

Scenario-Based Linux Interview Questions & Answers

1. User Management: A user complains they cannot log in. How will you troubleshoot?

1. Check if the user exists:
`cat /etc/passwd | grep username`
2. Verify the account is not locked:
`sudo passwd -S username`
3. Check failed login attempts:
`sudo faillog -u username`
4. Ensure correct permissions for the home directory:
`ls -ld /home/username`

2. File Permissions: A script is executable by one user but not another. How do you resolve this?

1. Check the script permissions:
`ls -l script.sh`
2. Add execute permission if missing:
`chmod +x script.sh`
3. Check file ownership:
`chown user:user script.sh`

3. Process Management: A service is consuming 100% CPU. How will you find and fix it?

1. Identify the process:
`top` or `htop`
2. Get detailed information:
`ps aux --sort=-%cpu`
3. Kill or restart the process:
`kill -9 PID` or `systemctl restart service`

4. SSH Issues: You cannot SSH into a remote machine. How do you debug?

1. Verify SSH service is running:
`sudo systemctl status sshd`
2. Check firewall rules:
`sudo iptables -L`
3. Ensure correct permissions for `~/.ssh/authorized_keys`
4. Look for errors in logs:
`sudo journalctl -u sshd --since '5 minutes ago'`

5. Disk Space Full: / partition is full. How do you find and delete large files safely?

1. Find large files:
`sudo du -ah / | sort -rh | head -20`
2. Delete safely:
`sudo rm -rf /path/to/large/file`
3. Check open deleted files:
`sudo lsof | grep deleted`

6. File Corruption: A log file is showing junk characters. How will you check and recover it?

1. Check encoding:
`file log.txt`
2. View non-printable characters:
`cat -v log.txt`
3. Try opening with vi or nano:
`vi log.txt`

7. Crontab Not Running: A scheduled job is not executing. How do you debug?

1. Verify crontab entries:
`crontab -l`
2. Check logs:
`sudo journalctl -u cron --since '5 minutes ago'`
3. Ensure correct permissions:
`chmod +x script.sh`

8. Package Installation Failing: yum or apt is failing. How will you resolve it?

1. Clean package manager cache:
`sudo apt clean` or `sudo yum clean all`
2. Check network connection:
`ping google.com`
3. Update package lists:
`sudo apt update` or `sudo yum update`

21. How do you extend a partition without unmounting it?

1. Identify the partition:
`lsblk`
2. Resize the filesystem (ext4 example):
`sudo resize2fs /dev/sdX`
3. If using LVM:
`sudo lvextend -r -L +10G /dev/mapper/vg-lv`

22. What steps are needed to add a new disk to a Linux server?

1. Identify the new disk:
`lsblk`
2. Create a partition:
`sudo fdisk /dev/sdX`
3. Format it:
`sudo mkfs.ext4 /dev/sdX1`
4. Mount it:
`sudo mount /dev/sdX1 /mnt`

23. How to check which processes are writing to a file in real time?

`sudo lsof +f -- /path/to/file`

24. How to recover a deleted file in Linux?

1. If the file is still open by a process:
`lsof | grep deleted`
2. Recover using extundelete (if on ext4):
`sudo extundelete /dev/sdX --restore-all`

25. How to check disk I/O performance in Linux?

1. Check disk activity:
`iostat -x 1`
2. Monitor real-time disk usage:
`iotop`

26. A directory is taking too much space. How do you analyze it?

`du -sh /path/to/directory/*`

27. How do you remount a filesystem in read-write mode without rebooting?

`sudo mount -o remount,rw /mnt`

28. What happens when a file is deleted but is still in use by a process?

The space is not freed until the process releases it.

Identify the process and restart it:

`lsof | grep deleted`

29. How do you create and mount a swap file?

1. Create a swap file:
`sudo fallocate -l 2G /swapfile`
2. Secure permissions:
`sudo chmod 600 /swapfile`
3. Set up swap:
`sudo mkswap /swapfile`
4. Enable swap:

```
sudo swapon /swapfile
```

5. Make it permanent:

```
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

30. How do you find large unused files across multiple partitions?

```
sudo find / -type f -size +1G -atime +30 -exec ls -lh {} +
```

31. How do you fix a corrupted file system using fsck?

1. Unmount the filesystem:

```
sudo umount /dev/sdX
```

2. Run fsck:

```
sudo fsck -y /dev/sdX
```

32. What is inode exhaustion, and how do you resolve it?

1. Check inode usage:

```
df -i
```

2. If inodes are full, delete small unused files.

33. You need to copy a huge file across servers. What's the fastest way?

1. Use rsync:

```
rsync -avz --progress file user@remote:/destination
```

2. Or scp with compression:

```
scp -C file user@remote:/destination
```

34. Your df -h and du -sh show different disk usage. Why?

Likely due to deleted files still held by a process.

Check with:

```
ls -l | grep deleted
```

35. How do you find out which directory is consuming the most in /var?

```
du -sh /var/* | sort -rh | head -10
```